

KICC DIGITAL ECONOMY PLATFORM

Refined Architecture Blueprint — Full Production Deployment

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1. Platform Vision

A unified global-facing Kenyan digital economy super-app serving 5 user roles:

 **VIEWER (Free — Global Population)**

- Personalized feed from SEO + search history + cross-platform social media signals
- Contact sellers directly or use platform escrow
- Browse 47 counties, products, travel, events, 3D/VR experiences
- Government services suggested contextually (Super eCitizen)
- No subscription required — monetized via ads + transaction fees

 **EXHIBITOR (Subscription — Tiered by Company Size)**

- Tiers: Starter (KSh 25k/yr) → Standard (KSh 60k) → Premium (KSh 120k) → Platinum (KSh 250k)
- Private portal — only exhibitor + KICC admin can see
- Post products/services for global visibility (B2C, B2B, G2C)
- Auto-generated promo videos from still images (AI video sampler)
- Content pre-approved by KICC or delegated authority (county gov)
- Add-ons: 3D walkthrough, VR experience, advertising campaigns
- Every sector of Kenyan economy supported — sectors dynamically managed in admin

 **KICC ADMIN (Super — Full Control)**

- Full authority over ALL counties + system-wide settings
- Organize & host MICE events (14+ venues: Tsavo Hall, Amphitheater, COMESA, Shimba Hills, VVIP Lounge, Courtyard, Lawn, Bilateral Rooms, Taifa Hall, etc.)
- 3D venue rendering — public spaces only (security classification restricts sensitive areas)
- Venue booking with cost & revenue-share calculations
- Approve/reject exhibitor & advertiser content

- Manage: visa rules, courier partners, payment gateways, subscription plans, pricing
- EVERYTHING adjustable via Filament admin — zero hardcoded configs

COUNTY GOVERNMENT (47 Counties — Each a Full Platform)

- Login → select county → personalized county hub
- Can personalize: sectors displayed, service pricing, content showcase
- Pay for services they subscribe to
- Sell county products directly (bulk & retail worldwide)
- Sponsor advertising for local small exhibitors
- Approve county-level content (if KICC has delegated authority)
- 14 devolved functions managed digitally: agriculture, health, transport, trade, planning, pre-primary education, environment, public works, firefighting, disaster management, drugs control, community participation, culture, natural resources
- Government structure: Governor → Deputy → CECM departments → Chief Officers → Sub-county → Ward → Village administrators

NATIONAL GOVERNMENT (21 Ministries + Agencies — NEW)

- 21 Ministries with 50+ State Departments, Judiciary, Constitutional Commissions
- Hierarchical permissions: Cabinet Secretary → Principal Secretary → Director → Officer
- Higher ranks inherit visibility of lower, not edit unless assigned
- Sell products/services directly (bulk & retail)
- Super eCitizen: contextual government service suggestions based on user location + behavior
- Specific integrations: Ministry of Lands (title verification), Ministry of Security (review validation), Ministry of Tourism (listing approval)

THIRD-PARTY VENUE (White-Label Package — NEW)

- Sarit Centre, Bomas of Kenya, etc. buy a venue package
- Get their own admin panel (white-labeled)
- Manage their own exhibitors, set pricing, ticketing
- KICC manages software + compliance + security
- Fixed package fee (no per-transaction royalties)
- Payment: merchant-direct OR KICC-handled (selectable)

2. Architecture Layers

Layer 1: Database — TiDB (MySQL-Compatible Distributed)

- **Production:** TiDB Dedicated cluster — auto-sharding, HTAP, multi-region
- **Backup:** 2 immediate-failover replicas (active standby)
- **Dev/Testing:** TiDB Serverless (upgraded as development progresses)
- **Data Sovereignty:** Kenyan data center for production (Data Protection Act 2019)
- Zero code changes between tiers — same Eloquent, same migrations
- TiFlash columnar replicas for real-time analytics without ETL

Layer 2: Application — Laravel 13

- Web Controllers (Blade SSR — SEO-optimized)
- API Controllers (REST — for 3D engine, mobile, third-party)
- Filament Admin (simplified UI per role: KICC super admin vs county admin vs exhibitor)
- Services: VideoGen, Payment, Shipping, AI Client, SMS, Multi-currency
- Queue workers (pipeline jobs, notifications, escrow triggers)
- Filament admin: EVERYTHING adjustable — plans, pricing, sectors, gateways, partners, fees, content

Layer 3: AI Pipeline & Automation

Tree-and-Branch Architecture — each branch independent, priority-ordered:

1. **Trade Branch** (1st priority — KICC mission: trade)
2. **Tourism Branch** (2nd priority)
3. **Government Services Branch** (Super eCitizen)
4. **County & National Content Branch**
5. **Advertising Branch** (video sampler, 3D, programmatic)
6. **Education & Culture Branch**

Each branch: Observer → Decider (LLM + rules) → Actor (n8n/APIs) → Learner (reinforcement)

Existing "monster" software used (n8n, Unreal/Unity) — no custom training models unless necessary

Layer 4: Frontend Delivery

- **Marketing/SPA:** Existing Cloudflare Worker + R2 (LottieFlow icons, county portal — kept & enhanced)
- **Application:** Laravel Blade SSR (SEO-optimized, full capability)
- **Three.js 3D:** Embedded in Blade pages for venue maps, county maps
- **Cloudflare:** Static assets, CDN, API caching, DDoS protection

Layer 5: Interactive 3D / Game Engine

- Best-in-class engine: Unreal Engine / Unity (KICC developers only)
- Connected as independent layer via REST APIs
- Security classification: public restricted classified
- PUBG/CoD-style interactive controls on mobile & desktop
- VR tourism, education visualizations, venue walkthroughs

3. Domain Model — All Tables

3.1 Core Platform (~70 tables)

Group	Tables	Notes
Users & Identity	users, viewer_profiles, exhibitor_profiles, kicc_admin_profiles, county_gov_profiles, national_gov_profiles, user_communication_channels	Auth isolated from profiles. Phone + Google login. MFA for gov accounts
Auth & Security	password_reset_tokens, sessions, phone_verification_codes, user_mfa_devices, oauth_clients, oauth_tokens	
Permissions	permissions, roles, model_has_permissions, model_has_roles, role_has_permissions	Spatie RBAC + hierarchical government inheritance
Government Hierarchy	government_levels, ministries, state_departments, government_agencies, government_officials, county_governments, county_executive_committee, county_departments, county_chief_officers, county_assembly, county_assembly_members, county_sub_counties, county_wards, county_villages, county_public_service_board, constitutional_commissions, commission_members, judiciary_bodies, judiciary_officials	NEW — full national + county structure
Counties & Sectors	counties, sectors, county_sector, seasonal_calendars, county_user	47 counties, dynamic sectors
Exhibitions	exhibitions, venues, event_sessions, booths, bookings, booking_booths, ticket_types, tickets	Physical + virtual + third-party
Subscriptions	subscription_plans, user_subscriptions, subscription_features, subscription_plan_features	Tiered pricing, all adjustable
County Entities	sector_entities, sector_entity_reviews, entity_trust_scores, county_tourism_attractions, county_hotels, county_products, county_institutions, county_farms, county_transport, county_health_facilities, county_culture_sites, entity_media, immersive_content_records	Per entity type, independent tables with government_level field
Audit & Logs	audit_logs, notification_logs, dispute_cases	Both TiDB + OpenSearch
Cache & Jobs	cache, cache_locks, jobs, job_batches, failed_jobs	

3.2 E-Commerce Marketplace (30 tables)

Group	Tables
Catalog	product_categories, products, product_variants, product_images, product_reviews, product_questions
Suppliers	suppliers, supplier_products
Inventory	warehouses, inventory, inventory_movements
Cart & Orders	shopping_carts, cart_items, orders, order_items, order_addresses, order_status_history
Fulfillment	shipments, shipment_items, shipment_tracking_events, return_requests

Wishlists & Coupons	wishlists, wishlist_items, coupon_codes, coupon_usages
Shipping	shipping_zones, shipping_rates, courier_partners, pickup_requests
Pricing	price_history

Features: Multi-currency (all currencies, real-time rates), B2C + B2B + G2C, Escrow for goods & services

3.3 Travel & Tourism (~32 tables)

Group	Tables
Flights	airlines, airports, flights, flight_inventory, flight_bookings, flight_passengers
Hotels	hotels, hotel_rooms, hotel_room_inventory, hotel_bookings, hotel_room_bookings
Restaurants	restaurants, restaurant_tables, restaurant_menu_items, restaurant_bookings
Transfers	airport_transfers, transfer_bookings, ride_hailing_partners
Travel	travel_packages, travel_package_items, travel_package_bookings, attractions, attraction_tickets, attraction_bookings, traveler_profiles, travel_itineraries, travel_itinerary_items, car_rentals, car_rental_bookings
Supporting	visa_rules (NEW), exchange_rates (NEW), gds_config (NEW)

3.4 AI Pipeline (~22 tables)

Group	Tables
Models & Embeddings	ml_models, model_versions, embeddings
Recommendations	user_segments, user_segment_assignments, match_scores, recommendations, recommendation_interactions
Pipeline	pipeline_branches (NEW), pipeline_jobs, pipeline_runs, pipeline_run_steps
Agentic Loop	agentic_loop_configs, agentic_loop_cycles, agentic_loop_logs
Experiments	a_b_tests, a_b_test_assignments, dynamic_pricing_rules
Content & SEO	content_generation_queue, seo_keyword_tracking
Operations	feature_flags

3.5 Advertising (18 tables)

Group	Tables
Campaigns	advertisers, ad_campaigns, ad_groups, ad_creatives, ad_placements, ad_targeting
Performance	ad_impressions, ad_clicks, ad_conversions, ad_budget_pacing
Programmatic	ad_programmatic_bids
Analytics	ad_analytics_daily
Payouts	ad_publisher_payouts
Moderation	ad_pre_approvals (NEW), video_sampler_jobs (NEW)
Billing	advertiser_payment_methods, advertiser_transactions

3.6 Payments & Subscriptions (16 tables)

Group	Tables
Gateways	payment_gateways
Transactions	payment_intents, transaction_logs, refunds, payment_disputes
Settlements	seller_payouts, settlement_batches, settlement_transactions
Billing	billing_cycles, invoices, invoice_items, tax_rates, currency_rates
Usage	usage_logs, payment_methods

3.7 SEO & Content (13 tables)

Group	Tables
CMS	content_pages, content_blocks, news_posts
SEO	seo_metadata, redirect_rules, sitemap_urls
Search	search_queries, tags, taggables
Analytics	page_views, analytics_events, page_speed_results, performance_kpis

3.8 3D Assets & Interactive Experiences (NEW — 8 tables)

Tables	Purpose
3d_scenes	Environments (county maps, venues, tourism sites)
3d_models	Individual objects (booths, buildings, rooms)
3d_walkthroughs	Guided paths through scenes
vr_sessions	User VR experience logs
interactive_experiences	Game-engine projects (Unreal/Unity)
experience_bookings	3D service orders by exhibitors/venues
3d_asset_versions	Versioning for models/scenes
asset_approvals	KICC moderation of 3D content

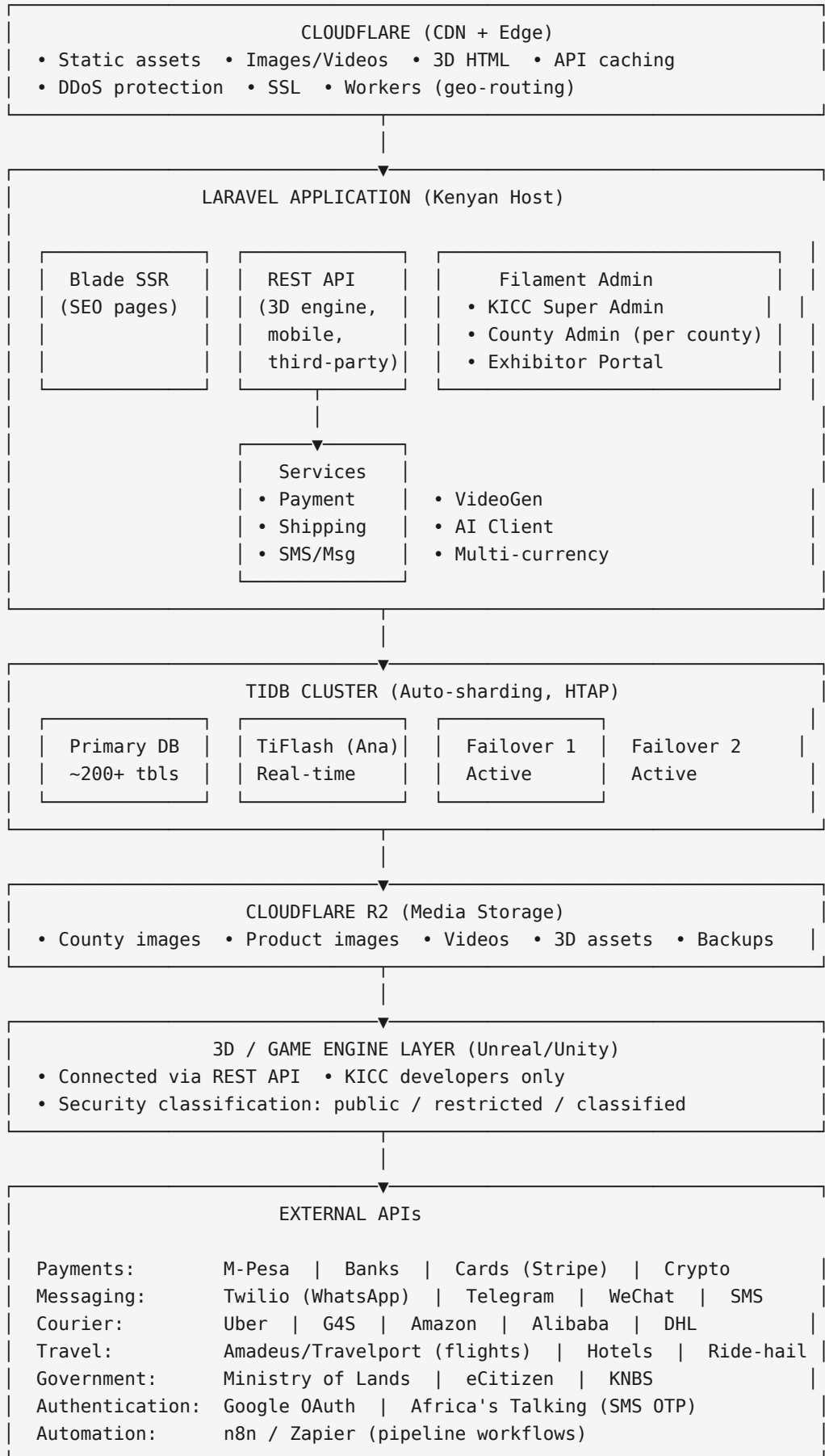
Security classification: public restricted classified

3.9 Communications (NEW — 6 tables)

Tables	Purpose
communication_channels	Master list (WhatsApp, Telegram, WeChat, SMS, Email, Russian apps)
user_channel_preferences	User → channel → identifier mapping
message_queue	Outbound messages pending send
message_logs	Sent messages with delivery status
message_templates	Reusable templates per channel
channel_providers	API config per provider (Twilio, Telegram bot, etc.)

One-way only. User replies → external support system (never stored in platform DB — SQL injection prevention)

4. Integration Architecture



5. Data Flow Examples (15 Scenarios)

5.1 Viewer (Free — Global Population)

1. User lands on platform (organic SEO / social ad / geo-targeted ad / direct)
2. AI Pipeline — SEO branch analyzes: location, device, season, weather, referrer, search query, cross-platform signals
3. Personalized landing page generated: winter user in Seattle → "Escape to Mombasa" / student in Nairobi → "47 counties"
4. User browses freely: county pages, exhibitor portfolios, travel, 3D/VR, government services (contextual suggestions)
5. Every action tracked → feeds AI Pipeline for better personalization
6. User finds something → Option A: Contact seller directly / Option B: Platform escrow (login required) / Option C: Book travel
7. Login via Google (preferred — enhances SEO) or Phone (SMS OTP — international provider)
8. Future visits: progressive personalization → builds trust → conversion

5.2 Tourist (International — Full Travel Booking)

1. Geo-targeted ad: "Visit Kenya — 28°C in Mombasa" → user clicks
2. Destination page shows: 3D/VR preview, packages, hotels, restaurants, visa info (auto-checked by nationality)
3. User selects destination → AI recommends: flights (GDS), hotels, restaurants, transfers
4. Book flight → PNR generated → passenger manifest → payment (card/crypto/M-Pesa)
5. AI auto-suggests: airport transfer (Uber assigned + driver details sent), hotel, restaurant aligned with arrival
6. Multi-stop itinerary builder: Mombasa → Tsavo → Nairobi (auto-books all segments)
7. Reminders via preferred channel (WhatsApp/Telegram/SMS/): 7 days → 24 hours → day of (driver name, plate)
8. Post-trip: rating & review prompted for every service
9. All listings pre-approved by Tourism (national + county) — quality control enforced

5.3 Exhibitor (Subscriber — Private Portal)

1. Register → select tier (Starter/Standard/Premium/Platinum) → pay → portal activated
2. Private portal: dashboard (views, inquiries, revenue, ratings), product management, media upload, communications
3. Upload product images → AI video sampler generates promo video automatically
4. Content → moderation queue → KICC/county approval → live
5. AI generates embedding vector + SEO metadata via n8n
6. Product visibility determined by: trust_score + review_rating + recency (fairness via rating, not paid placement)
7. Viewer inquiries: direct contact OR platform escrow

8. Add-ons: book physical/virtual booths, order 3D visualization, buy ads, B2B matchmaking
9. Tier limits enforced by system (usage_logs) — upgrade to unlock

5.4 KICC Admin

1. Login with official account → MFA → admin dashboard
2. Modules: Exhibition Management (14+ venues, 3D rendering public-only), Content Moderation (approve/reject/delegate), Subscriptions & Plans (create/edit/delete), Payment Gateways (add/edit M-Pesa/Stripe/crypto/banks, set fees, handle escrow disputes), Courier Partners (add/edit Uber/G4S/Amazon, configure APIs, set zones/rates), Travel & Tourism (visa rules, GDS creds, approve listings), Government Relations (map officials, set hierarchical permissions), Advertising (placement slots, approve campaigns, set pricing), 3D & Immersive (manage scene library, security classification), System (audit logs, pipeline jobs, feature flags, user management, KPIs)
3. EVERYTHING adjustable — no hardcoded configs

5.5 County Government

1. Login → select county → county hub
2. Personalize: sectors displayed, service pricing, content showcase
3. County profile: hero image, tagline, economic data, population, climate, seasonal calendar, tourism highlights
4. Manage: tourism attractions, hotels, products, institutions, farms, transport, health facilities, culture sites
5. Sell products directly (bulk & retail worldwide) — payment via escrow or direct, courier picks up from county depot
6. Sponsor advertising for local small exhibitors
7. Approve county-level content (if KICC delegated)
8. Analytics: who bought, from where, revenue
9. CEDP departments reflected in permissions structure

5.6 National Government

1. Login → ministry/agency dashboard (permissions based on position in hierarchy)
2. Higher ranks inherit visibility of lower, not edit unless assigned
3. Sell products/services directly (bulk & retail)
4. Super eCitizen: platform detects user context → queries government service DB → suggests relevant services as contextual links ("Need a business permit? Mombasa County Trade Dept")
5. Ministry of Lands: land title verification API
6. Ministry of Security: review & rating validation
7. Ministry of Tourism: approves travel/hospitality listings

5.7 Escrow Flow (Goods & Services)

Goods: Buyer pays → platform holds → quality inspection → courier picks up → tracking events flow → buyer confirms → funds released to seller minus commission. Dispute → frozen, KICC mediates via dispute_cases.

Services: Book → receive ticket/voucher immediately → pay. No hold — service delivered at booking.

5.8 B2B Flow

Buyer → search → AI recommends suppliers → RFQ → seller responds with bulk pricing → contract generated → escrow payment → bulk courier → both parties rate each other.

5.9 Third-Party Venue

Venue buys package → white-label admin panel → manage own exhibitors, pricing, ticketing → KICC handles software + compliance + security → fixed package fee (no royalties).

5.10 Private Individual (Landlord/Property Seller)

List property with images + exact location → Ministry of Lands verifies title → credibility scored → buyers view + see validated neighborhood reviews → contact seller or use escrow → legal transfer per Kenyan property law → bank financing optional.

5.11 Super eCitizen

System watches user behavior + location + profile → queries government service DB → inserts contextual suggestion links in UI ("Based on your location in Mombasa, here are 3 county services you may need").

5.12 AI Pipeline Tree

6 branches, priority-ordered: Trade (1st) → Tourism (2nd) → Government Services (3rd) → County/National Content (4th) → Advertising (5th) → Education & Culture (6th)

Each branch: Observer ↔ Decider (LLM + rules) ↔ Actor (n8n/APIs) ↔ Learner (reinforcement)

All run independently, conjoined at master pipeline level. Existing software used — no custom training unless necessary.

5.13 Advertising Pipeline

Upload images → AI video sampler (priority queue) → 3D model (high-value) → 4K for big screens/VR → campaign created with targeting → KICC approval → live (web + big screens + VR) → performance tracked → AI optimizes → programmatic bidding for premium slots.

5.14 Education Institution

Subscribe → access 3D library: Mt. Kenya VR climb, Menengai crater flyover, Maasai Mara migration, Lake Nakuru flamingos → interactive controls (walk/climb/look) → curriculum-matching reports for teachers.

5.15 Communications (One-Way)

System event → user's preferred channel (WhatsApp/Telegram/WeChat/SMS/Email) → message queued → provider API called → delivered → logged. User replies → external support (never in platform DB — SQL injection prevention).

6. Deployment Strategy

Single Phase — Full Production (All-in)

No staged rollout. Everything deploys at once:

- Laravel 13 on Kenyan-hosted servers (data sovereignty per Data Protection Act 2019)
- TiDB Dedicated cluster (auto-sharding, HTAP, 2 active failover replicas)
- All ~240+ tables across 9 domains deployed with initial seed data (47 counties, 21 ministries, pricing plans, sectors, etc.)
- 3D Engine layer operational (Unreal/Unity — KICC developers only)
- All payment gateways live: M-Pesa, banks, cards (Stripe), crypto, alternatives
- All courier partners integrated: Uber, G4S, Amazon, Alibaba, KICC-approved
- All 6 AI pipeline branches running (Trade, Tourism, Gov, Content, Advertising, Education)
- Super eCitizen contextual suggestions live across platform
- Multi-currency with real-time exchange rate feeds
- Full government hierarchy populated (national + 47 counties + judiciary + commissions)
- Cloudflare CDN + DDoS for static assets and API caching
- Admin portal: KICC super admin + 47 county admins with full personalization
- Filament admin: every setting, price, plan, partner, and policy adjustable without code

Recommended Hosting Stack (Testing)

Component	Testing (Current)	Production (Future)
Application	Laravel local (php artisan serve)	Kenyan data center (EADC/iColo/Safaricom Cloud)
Database	TiDB Cloud (MySQL protocol)	TiDB Dedicated + 2 failover replicas
Media Storage	Cloudflare R2	Cloudflare R2 + Kenyan backup
3D Engine	Local Unreal/Unity builds	GPU server (Hetzner/Dedicated)
CDN	Cloudflare (free)	Cloudflare (enterprise)
Auth	Laravel Sanctum + Google OAuth + SMS OTP	Same + MFA for all gov accounts

7. Project Structure

```

/home/kicc/Desktop/kicc/kicc-platform/
├── app/
│   ├── Console/Commands/      Artisan commands
│   ├── Http/
│   │   ├── Controllers/
│   │   │   ├── Api/          REST API (Auth, Booking, Booth, County,
│   │   │   │               Exhibition, Pipeline, Ticket, Venue, Travel...)
│   │   │   └── Web/          Blade SSR (Home, Counties, Exhibitions,
│   │   │                       Dashboard, Venue, Screen, Social Auth...)
│   │   └── Middleware/      AgenticSEO, auth, security
│   ├── Models/              32+ Eloquent models
│   ├── Services/            VideoGen, Payment, Shipping, SMS, AI Client
│   ├── Filament/            Admin panel resources (all adjustable)
│   └── Providers/           AppServiceProvider
├── config/
│   ├── database.php          TiDB + SQLite
│   ├── screens.php
│   └── ...                   All Laravel configs
├── database/
│   ├── schema.sql            Core Platform (56 tables)
│   ├── schema-commerce.sql   E-Commerce (30)
│   ├── schema-travel.sql     Travel & Tourism (28)
│   ├── schema-ai-pipeline.sql AI Pipeline (20)
│   ├── schema-advertising.sql Advertising (15)
│   ├── schema-payments.sql   Payments & Subscriptions (16)
│   ├── schema-seo-content.sql SEO & Content (13)
│   ├── schema-full.sql       Loads all above
│   ├── migrations/          Laravel migration files
│   └── seeders/              Initial data seeders
├── public/
│   ├── 3d/                   Three.js experiences
│   └── index.php
├── resources/views/
│   ├── counties/             County pages (47)
│   ├── exhibitions/          Exhibition listings & booking
│   ├── travel/               Travel booking UI
│   ├── dashboard/            Exhibitor portals
│   ├── auth/                 Login, register, verify
│   ├── layouts/              App shell, SEO headers
│   └── screens/               Screen display
├── routes/
│   ├── web.php               Browser routes (Blade SSR)
│   ├── api.php               REST API routes
│   └── console.php            Scheduled tasks
├── storage/app/public/        County images, videos, 3D assets
└── setup.sh                  One-command setup

```

└─ composer.json	Laravel 13 + Filament + Sanctum + Socialite + Cashier
└─ BLUEPRINT.md	This document (Markdown)
└─ BLUEPRINT.pdf	This document (PDF)
└─ README.md	

Key Dependencies (composer.json)

Package	Purpose
laravel/framework ^13.8	Core framework
laravel/sanctum ^4.0	API tokens (millions of users)
laravel/socialite ^5.29	Google login (primary auth method)
laravel/cashier ^15.0	Subscription billing
filament/filament ^5.7	Admin panel (everything adjustable)
spatie/laravel-permission ^8.3	RBAC + hierarchical permissions
africastalking/africastalking ^3.0	SMS (international provider)
league/flysystem-aws-s3-v3 ^3.35	Cloudflare R2 storage